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SE3071: DIGITAL IMAGE PROCESSING - 50% PROGRESS REVIEW
VIVA PREPARATION GUIDE & SYSTEM SPECS
DOMAIN: SECURITY / SURVEILLANCE

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1. PROJECT OVERVIEW & METADATA

- Module: SE3071 - Digital Image Processing (Year 3 Semester 2 - 2026)
- Allocated Domain: Security / Surveillance (Based on IT Number Last Digit: 6)
- System Architecture: Single Integrated Python App (Streamlit + OpenCV + NumPy)

2. PROBLEM STATEMENT

"Enhancing low-contrast surveillance images and detecting candidate vehicle, person, and license plate regions using adaptive thresholding and morphological contour analysis."

3. DATASET SUMMARY

- Total Images: 30 Images
 - * Self-Captured Images: 10 images (Roadway traffic, scooters, vehicles, and pedestrians)
 - * Downloaded Images: 10 images (CCTV surveillance frames, low-light traffic scenes)
 - * Unseen Test Dataset: 10 images (Held back exclusively for quantitative evaluation)
- Documentation File: dataset_record.txt (Includes image names, resolution, format, and test purpose)

4. CORE PROCESSING PIPELINE & VIVA JUSTIFICATIONS

Q1: What are the Colour Spaces used and why?

Answer:

- BGR to Grayscale: Essential for noise reduction, enhancement (CLAHE), and edge detection algorithms.
- BGR to HSV: Preserves color intensity independent of brightness, useful for reflective region identification like vehicle license plates or tail lights.

Q2: How did you perform Contrast Enhancement and Illumination Correction?

Answer:

- Contrast Enhancement: Applied CLAHE (Contrast Limited Adaptive Histogram Equalization). Unlike standard Histogram Equalization, CLAHE operates on localized regions (tiles) and clips high peaks to prevent noise over-amplification in surveillance scenes.
- Illumination Correction: Implemented Gamma Correction via Look-Up Table (LUT). This brightens dark shadowed areas without saturating well-lit foreground zones.

Q3: Which Noise Reduction filters did you compare, and which one did you select?

Answer:

- Comparison: Compared Gaussian Blur vs Median Blur.
- Selection Justification: Selected MEDIAN BLUR. Median filtering is a non-linear

r method exceptionally effective at removing salt-and-pepper noise (typical in low-light CCTV sensor artifacts) while strictly preserving sharp edges necessary for object bounding box detection.

Q4: Why did you choose Adaptive / Dynamic Thresholding over Otsu's Thresholding?
Answer:

- Justification: Surveillance images often suffer from uneven spatial illumination (e.g., street lights, headlight glare, tree shadows). Otsu's method calculates a single global threshold, which fails under non-uniform lighting. Adaptive Thresholding computes threshold values for smaller local regions, adapting better to real-world security scenes.

5. DOMAIN ANALYSIS & OUTPUT RULE LOGIC

- Domain Techniques Implemented:

1. CLAHE & Gamma Illumination Enhancement
2. Median Noise Filtering
3. Canny Edge Detection
4. Morphological Closing (to connect fragmented edge components)
5. Contour Analysis & Geometric Feature Bounding Boxes

- Output Decision Logic (Calculation):

"Contours are filtered based on bounding rectangle area (area > 300 pixels) and aspect ratio ($0.2 < \text{width/height} < 6.0$). Matching regions are rendered with green bounding boxes, and the system outputs the total count of candidate objects."

6. EVALUATION METRICS & GROUND TRUTH

- Evaluation Metrics Selected:

1. Count Error: $|\text{Actual Manual Count} - \text{System Detected Count}|$
2. Processing Time per Frame (seconds)

- Ground Truth Preparation: Manual object counting log established for the 10 unseen test set images.

7. VIVA CHANGE PREDICTION DRILLS (IF EXAMINER ASKS TO CHANGE PARAMETERS)

- Scenario 1: "Increase Canny High Threshold"

-> Effect: Edge detection becomes stricter. Minor noise edges disappear, but faint object boundaries might be missed.

- Scenario 2: "Decrease Gamma Value (below 1.0)"

-> Effect: Image gets darker, low-light CCTV regions become less visible, reducing contour detection yield.

- Scenario 3: "Increase CLAHE Clip Limit"

-> Effect: Contrast increases in local tiles, making faint edges more prominent, but may introduce localized noise.

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